

DIELECTRICS

1. What are Dielectrics?

Dielectrics are insulating material in which all the electrons are bound to their parent molecules. They have permanent electric dipoles or have an ability to produce enormous induced dipoles in the presence of external electric field.

2. Define dielectric constant.

Dielectric constant is the measure of the polarization produced in the material. It is the ratio between the absolute permittivity (ϵ) and the permittivity of free space (ϵ_0)

$$\epsilon_r = \epsilon / \epsilon_0$$

3. Define Polarization of a dielectric material?

The process of producing electrical dipoles inside the dielectric by an external electric field is called polarization in dielectrics.

4. Define Polarization vector and Electric displacement vector.

Polarization vector (P): It is defined as the dipole moment per unit volume

$$P = \mu / V$$

Where μ - is the dipole moment

V- Volume of the specimen

Electric Displacement Vector (D): It is the amount of electric flux or energy per unit area.

$$D = Q/A$$

Where Q- amount of flux or energy

A- Area of the dielectric

5. Prove that $\chi_e = \epsilon_r - 1$

We know that Polarisation vector $P = \epsilon_0 \chi_e E$

or $\chi_e = P / \epsilon_0 E$

since $P/E = \epsilon_0 (\epsilon_r - 1)$

$$\chi_e = \epsilon_0 (\epsilon_r - 1) / \epsilon_0$$

$$\chi_e = (\epsilon_r - 1)$$

6. Name the four polarization mechanisms.

The four types of polarization mechanisms are

- (i) Electronic Polarization
- (ii) Ionic Polarization
- (iii) Orientational Polarization
- (iv) Space-Charge Polarization

7. What is meant by Electronic Polarization?

Electronic Polarization means production or inducement of electric dipoles by the applied electric field. It is due to shifting of charges in the material by the applied electric field.

8. What is meant by Ionic Polarization?

Ionic polarization is due to the displacement of cations (negative ions) and anions (Positive ion) in opposite direction due to application of an electrical field. This occurs in an ionic solid.

9. What is meant by Orientation Polarisation?

When an electric field is applied on molecules which possess permanent dipole moment, they tend to align themselves in the direction of applied field. The polarization due to such an alignment is called Orientation Polarization.

10. Explain Space-Charge Polarization?

When a dielectric is placed in an electric field the diffusion of ions take place along the field direction, giving rise to redistribution of charges in the dielectrics and this phenomenon is known as space charge polarization.

11. How does the dielectric polarization in polar and non-polar substances vary with temperature?

(OR)

What are the difference between Polar and Non-Polar molecules?

Sl.No	Polar Molecules	Non-Polar Molecules
1	These molecules have permanent dipole moments even in the absence of an applied field.	These molecules do not have permanent dipole moments.
2	The polarization of polar molecules is highly temperature dependent.	The polarization of this kind of molecules is independent of temperature.
3.	These molecules do not have symmetrical structure and they do not have centre of symmetry.	These molecules have symmetrical structure and they have centre of symmetry
4.	For this kind of molecules, there is absorption or emission in the infrared range.	For these molecules, there is no absorption or emission in the range of infrared.
5	Examples: CHCl_3 , H_2O , HCl	Examples: CCl_4 , CO_2 , H_2

12. Define Local or Internal field in dielectrics.

When a dielectric is kept in an external electric field (E), two fields are exerted due to (i) External field and (ii) dipole moment created. This long range or coulomb force which is created due to the dipoles is called internal field or local field in dielectric. It is given by $E_{int} = E + P / 3\epsilon_0$

Where P- Polarization

ϵ_0 – Permittivity in free space

13. What is meant by dielectric loss and loss tangent? Why it occurs?

When a dielectric material is subjected to electric field, the electrical energy is absorbed by the dielectric and certain amount of electrical energy is dissipated in the form of heat energy. This loss in energy in the form of heat is called dielectric loss.

The Power loss $PL \propto \tan \delta$, where $\tan \delta$ is called loss tangent and δ is called loss angle.

This dielectric loss mainly occurs due to the imaginary part of complex dielectric constant.

14. What is meant by dielectric breakdown and dielectric strength?

Dielectric strength: When external field applied to a dielectric material is greater than the critical field, the dielectric loses its insulating property and becomes conducting. Therefore a large current flow through the material. This phenomenon is called dielectric breakdown.

Dielectric Strength: It is the minimum strength (voltage) required per unit thickness of the dielectric material to produce dielectric breakdown.

15. Distinguish between Lorentz force and Coulomb force in dielectrics.

Lorentz force: It is the repulsive force between the nucleus and the electron cloud of a dielectric material when kept in external electric field.

Coulomb force: It is the attractive force between the nucleus and the electron cloud, which tries to maintain the equilibrium position of the dielectric material, kept in an external electric field.

16. What is meant by Ferro-electricity, Piezo-electricity and Pyro electricity?

Ferro-electricity: When a dielectric material acquires electric polarization even in the absence of external field, it is known as ferro electricity and these materials are termed as ferro-electrics

Piezo-electricity: When a dielectric material acquires electric polarization due to the external mechanical pressure, it is called piezo electricity and these materials are termed as piezo-electrics.

Pyro electricity: When a dielectric material acquires electric polarization due to thermal energy applied externally, it is called Pyro electricity and these materials are termed as Pyro electrics.

17. What is the effect of frequency of an a.c. field on polarization?

In general the polarization decreases with the increase in frequency. At optical frequencies electronic polarization is present and at infrared frequencies the ionic polarization occurs. The orientation and space charge polarization will be absent at optical frequencies and will occur only at radio and audio frequencies respectively.

18. What is the effect of temperature on Polarization?

- (i) The electronic and ionic polarization are independent of temperature
- (ii) The orientation polarization decreases with the increase in temperature whereas the space charge polarization increases with the increase in temperature.

19. What are different break down mechanisms that occur in a dielectric material?

The following are the break down mechanisms that occur in dielectrics

- (i) Intrinsic or Avalanche breakdown
- (ii) Thermal breakdown
- (iii) Chemical breakdown
- (iv) Discharge breakdown
- (v) Defect breakdown

20. Give any four properties of a Ferro electric materials

- (i) Ferro electric material will have permanent dipole moments
- (ii) They acquire polarization spontaneously even at weak fields
- (iii) They have Ferro-electric domain structure
- (iv) These materials behaves as para electric materials above Curie temperature

21. List out any four applications of Ferro electric materials

- i) Ferro electric materials are used in converting thermal energy into electrical energy
- (ii) They are useful in modulating the optical signals
- (iii) Electrets a type of Ferro electric material is used in recording unit, relay switches etc.
- (iv) In medicine it improves the tensile strength of the wound

22. Give any four examples of Ferro-electric materials

- (i) Rochelle salts
- (ii) Barium Titanate
- (iii) Potassium dihydrogen phosphate
- (iv) Lithium Niobate